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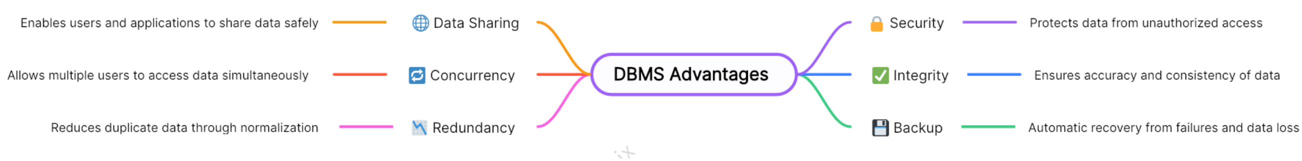
1. Comparison Assignment: Flat File Systems vs. Relational Databases

	Flat File Systems	Relational Databases

Structure	Stores data in a single file or multiple separate text files such as TXT file. Each record is typically a line, and fields are separated by commas, or tabs.	Organizes data into multiple related **tables**. Each table represents an entity, and relationships are established using **keys** (primary and foreign).
Data Redundancy	High –> because there is no way to ensure relationships or consistency, the same data may appear in several files or records.	Low –> to minimize duplication and guarantee data integrity, data is normalized and kept in distinct tables.
Relationships	no explicit backing for dataset relationships. Relationships must be managed manually using file processing or programming.	Supports complex relationships: - one-to-one - one-to-many - many-to-many through **foreign keys** and **joins**.
Example Usage	- Small applications - Simple data storage - Configuration files	- Banking, e-commerce - Hospital management
Drawbacks	- Hard to maintain as data volume increases. - A high danger of redundancy and inconsistency. - Complex queries are challenging to execute. - Scalability and security are limited.	-Needs setup and database management software. - More difficult to design. - For large systems, more processing power can be needed.

2. Database Management System (DBMS) Advantages – Mind Map



3. Roles in a Database System

- System Analyst:

Role: Serve as a link between technical solutions and business needs.

Responsibilites:

- Studies the organization's requirements and determines what the database system must do.

- Evaluates current systems and pinpoints areas in need of development.

- Creates system specifications that guide database and software design.

- Ensures that the finished system satisfies corporate objectives by collaborating closely with developers, database designers, and end users.

- Database Designer:
  Role: Designs the structure and organization of the database.
  Responsibilities:
  - Creates the **data model** (conceptual, logical, and physical designs).
  - Defines tables, relationships, keys, and constraints.
  - Ensures the database structure supports data integrity and performance.
  - Works with system analysts and developers to align design with application needs.

- Database Developer:
  Role: Builds and implements the actual database based on the design.
  Responsibilities:
  - Writes **SQL scripts**, **stored procedures**, and **triggers**.
    - Implements the database schema and data relationships.
    - Optimizes queries for performance.
    - Tests and debugs database components before deployment.

- Database Administrator (DBA):
  Role: Maintains and secures the database system after deployment.
  Responsibilities:
    - Manages **user access**, **security**, and **permissions**.
    - Performs **backups**, **recovery**, and **performance tuning**.
    - Monitors database health and ensures availability.
    - Applies patches, updates, and ensures data integrity.

- Application Developer:
  Role: Builds the software or applications that interact with the database.
  Responsibilities:
    - Develops **front-end interfaces** and **business logic** to connect to the database.
    - Uses programming languages (e.g., Java, Python, C#, PHP) to perform CRUD operations (Create, Read, Update, Delete).
    - Works closely with database developers to ensure smooth data flow between the app and database.

- BI (Business Intelligence) Developer:
  Role: Turns raw data into meaningful insights for decision-making.
  Responsibilities:
    - Designs **data warehouses** and **data pipelines**.
    - Creates **reports**, **dashboards**, and **visualizations**.
    - Works with stakeholders to identify key performance indicators (KPIs).
    - Ensures data is accurate, timely, and well-presented for business analysis.

4. Types of Databases

4.1 Relational vs. Non-Relational Databases

4.1.1 Relational Databases (RDBMS)

Description:

- Store data in **tables (rows and columns)** with predefined schemas.
- Use **SQL (Structured Query Language)** for querying and managing data.
- Support **relationships** between tables using **primary** and **foreign keys**.

Examples: MySQL, Oracle Database, PostgreSQL, Microsoft SQL Server.

Use Cases:

- Banking systems
- Hospital management systems
- E-commerce order tracking

4.1.2 Non-Relational Databases (NoSQL)

Description:

- Store data in **flexible formats** such as documents, key-value pairs, graphs, or wide columns.
- Do **not require a fixed schema**, allowing easy scaling and handling of unstructured data.
- Designed for **high performance** and **large-scale data**.

Examples:

- **MongoDB:** Document-based (stores data as JSON-like documents).
- **Cassandra:** Wide-column store, ideal for large, distributed datasets.

Use Cases:

- Social media platforms (storing user activity data)
- Real-time analytics
- IoT (Internet of Things) applications

4.2 Centralized vs. Distributed vs. Cloud Databases

	Centralized	Distributed	Cloud Databases
Description	- All data is stored in one central location (e.g., one server). - Users from different locations access data through a network.	- Data is stored across multiple locations or servers but appears as a single database to users. - Improves reliability, speed, and availability	- Hosted on cloud platforms like AWS, Azure, or Google Cloud. - Managed and accessed through the internet.
Advantages	- Easier to manage and back up. - Centralized control improves consistency.	- Fault tolerance and improved performance. - Data can be processed closer to where it's needed.	- Highly scalable, cost-effective, and accessible from anywhere. - Providers handle maintenance, backup, and security.
Drawbacks	Single point of failure — if the server goes down, the database becomes unavailable.	- Higher Cost - Slower Query Performance	- Internet Dependency - Downtime Risk
Examples	A company's internal HR database hosted on a single on-site server.	Google Spanner, Apache Cassandra.	Amazon RDS, Google Cloud SQL, Microsoft Azure SQL Database.

Use Cases	Small to medium organizations with limited data users.	- Large organizations with global operations (e.g., multinational companies). - Applications requiring fast local data access from multiple regions.	- Web applications - Mobile apps

5. Cloud Storage and Databases

5. 1 Cloud Storage

Cloud storage is a service that allows users and organizations to store, manage, and access data over the internet instead of using local drives or on-premises servers.

Examples: Google Drive, Amazon S3, Microsoft Azure Blob Storage.

How It Supports Databases:

- Provides the infrastructure for storing and managing database data.
- Ensures scalability, high availability, and data backup for cloud-hosted databases.
- Enables distributed access with minimal latency worldwide.
- Supports automatic replication, data recovery, and long-term persistence for cloud databases such as **Azure SQL** or **Amazon RDS**.

5.2 Advantages of cloud-based databases

1. **Accessibility:** Available from anywhere with an internet connection.
2. **Scalability:** Easily scale storage and computing power as needed.
3. **Security:** Offers encryption, access controls, and compliance features.
4. **Backup and Recovery:** Automated backup and recovery options in case of failures.
5. **Integration:** Seamlessly integrates with other cloud services.

5.3 Disadvantages of cloud-based databases

1. **Internet Dependency:** Databases are inaccessible without an internet connection.
2. **Downtime Risk:** Service outages from the provider can cause temporary unavailability.
3. **Security & Privacy Concerns:** Data is stored on third-party servers — introducing potential risks of unauthorized access or data breaches.